

Official YIMO 2 Scoring Guide

Let N_P denote the total number of contestants in the preliminary round. For each preliminary problem i , let $C_{p,i} \in \{0, 1\}$ denote contestant p 's score, and define

$$S_i = \sum_{p=1}^{N_P} C_{p,i}. \quad (1)$$

The weight of preliminary problem i is

$$f(i) = \min \left\{ 6, 1 + \ln \left(\frac{N_P + 1}{S_i + 1} \right) \right\}. \quad (2)$$

Let N_F denote the total number of contestants in the final round. For each proof problem i , let $P_{p,i} \in \{0, 1, \dots, 7\}$ denote contestant p 's score, and define

$$T_i = \sum_{p=1}^{N_F} \frac{P_{p,i}}{7}. \quad (3)$$

The weight of final proof problem i is

$$g(i) = \min \left\{ 6, 1 + \ln \left(\frac{N_F + 1}{T_i + 1} \right) \right\}. \quad (4)$$

For each final computational problem i , let $Q_{p,i} \in \{0, 1\}$ denote contestant p 's score, and define

$$U_i = \sum_{p=1}^{N_F} Q_{p,i}. \quad (5)$$

The weight of final computational problem i is

$$h(i) = \min \left\{ 6, 1 + \ln \left(\frac{N_F + 1}{U_i + 1} \right) \right\}. \quad (6)$$

For an individual contestant, let C_i , P_i , and Q_i denote their scores on the corresponding preliminary, proof, and computational problems. Define the final-round participation indicator by

$$I_F = \begin{cases} 1, & \text{if the contestant participated in the final round,} \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

The contestant's overall score is

$$R = \frac{20 \sum_{i=1}^{20} C_i f(i)}{\sum_{i=1}^{20} f(i)} + I_F \frac{20 \left(\sum_{i=1}^2 \frac{P_i}{7} g(i) + \sum_{i=1}^2 Q_i h(i) \right)}{\sum_{i=1}^2 g(i) + \sum_{i=1}^2 h(i)}. \quad (8)$$

If the contestant did not participate in the final round, their final-round contribution is defined to be 0.